

Average Trap Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	average-trap

Big idea: An average is a choice of representation, not a neutral summary of reality.

Overview

Students compare mean, median, and mode for skewed datasets and evaluate which average supports the most responsible claim.

Student Prompt

Inspect Average Trap Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Average Trap Stimulus A	Student-facing stimulus 1: Revision hours: 1, 1, 2, 2, 3, 3, 18. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Average Trap Stimulus B	Student-facing stimulus 2: Screen time claim: mean looks high, median looks moderate, mode shows the most common routine. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Average Trap Reveal / Twist	Student-facing stimulus 3: Screen time claim: mean looks high, median looks moderate, mode shows the most common routine. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — An average is a choice of representation, not a neutral summary of reality.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Average Trap Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Average Trap Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does choosing a summary statistic shape knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how an average is a choice of representation, not a neutral summary of reality.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Choose skewed datasets where one large value pulls the mean away from most cases.
- Ask students which number they would put in a headline before calculating all averages.

Ready-to-Use Examples

- Revision hours: 1, 1, 2, 2, 3, 3, 18.
- Screen time claim: mean looks high, median looks moderate, mode shows the most common routine.

Suggested Timing

Stage	Teacher move	Watch for
1	Give students a skewed dataset and ask them to describe a typical value.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups calculate or identify the mean, median, and mode.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students match each average to a possible headline or policy claim.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups decide which average is most responsible for the audience and purpose.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why one correct number can create a partial picture.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- All three averages can be correct; the TOK question is which one responsibly represents the situation.
- What does this number make visible, and what does it hide?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Support: give pre-calculated averages.
- Challenge: ask students to design a dataset where the three averages tell three different stories.

Extension Tasks

- Students find a public average and ask whether mean, median, or mode would be more responsible.
- Compare average choice with graph scale choice.

Related Activities

- Simpson's Paradox Shuffle
- Log Scale Surprise
- Graph Caption Framing